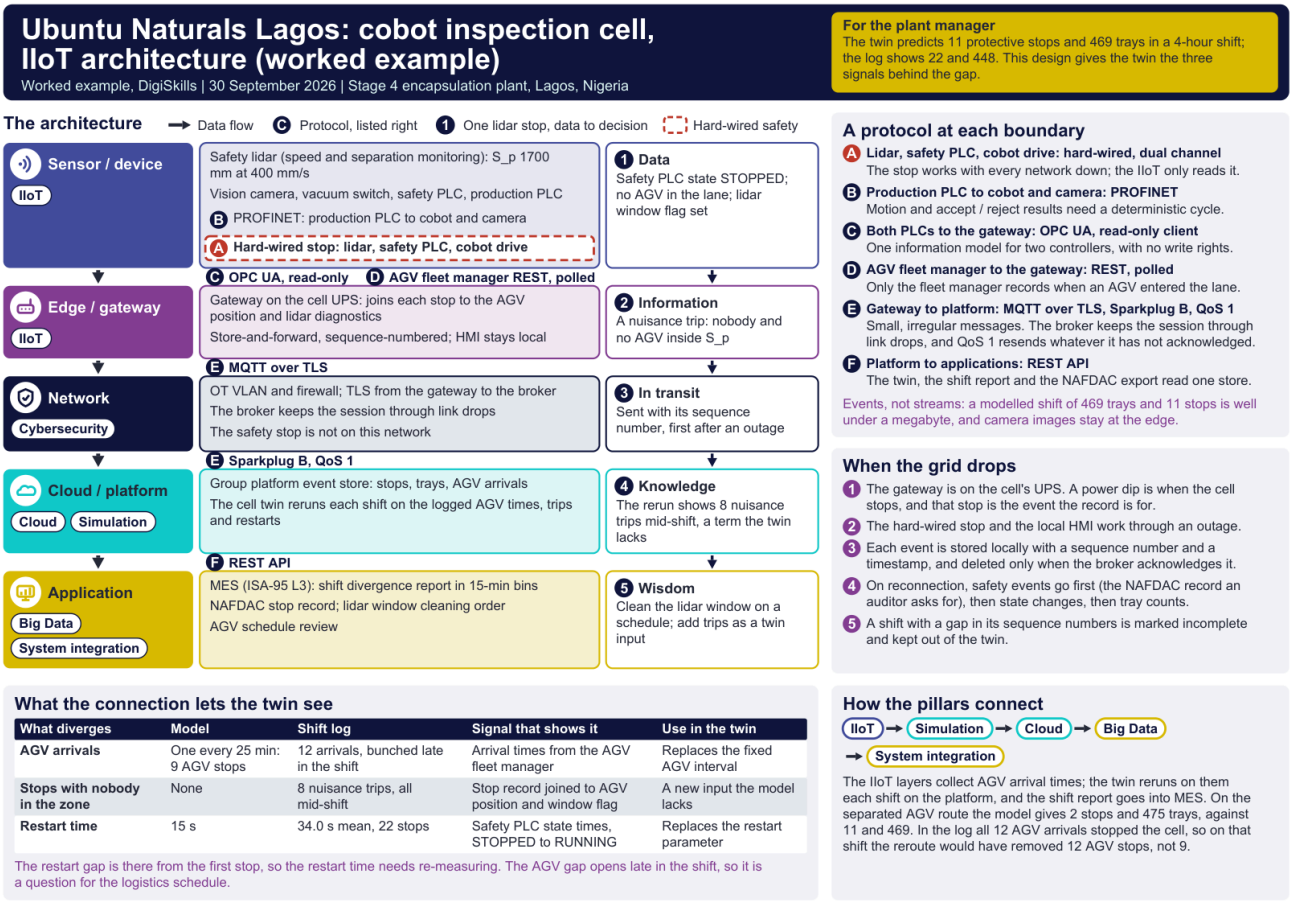
A04 worked exemplar: IIoT architecture poster for the Lagos cobot cell

This is a worked example of an A04 submission, made by the programme team to show a poster at the upper bands. It is set at the polishing / inspection cobot cell in Ubuntu Naturals' Lagos encapsulation plant, the cell from the W11 master class, and not at the Antananarivo baobab plant your brief describes. Its sensors and protocols belong to a robot cell, not a drying tunnel, so none of them answers your brief. Do not copy it. The Merit and Distinction bands are awarded for fitting the architecture to Antananarivo and the Tier-3 family, and borrowed layers or wording answer the wrong plant.

The poster

The submission is `A04_exemplar_lagos_cell_poster.pdf`, an A1 landscape sheet with no text smaller than 24 pt at print size. The preview below is scaled down; a marker reads the PDF at 100%.



The W11 twin of the Lagos cell predicts 11 protective stops and 469 trays in a 4-hour shift, and the plant's log for one shift records 22 stops and 448 trays. The poster designs the data connection that lets the twin see where that difference comes from.

How it maps to the marking scheme

Rubric row	Band it aims at	Why	What would lift it a band
Architecture Design	Distinction	The five layers hold the cell's own equipment, and Lagos conditions change the design: the grid puts the gateway on the cell's UPS, NAFDAC sets the order the buffer empties in, and the safety stop stays off the network.	Showing how the same sheet serves a second cell or the Johannesburg line, saying which parts are per cell and which are shared.
Sensor and Protocol Specification	Distinction	Every boundary has a protocol and a reason tied to the cell, determinism is handled by keeping the stop and motion commands off the IIoT path, and data volume is estimated from the model's counts.	A protocol comparison matrix with data rates and payload sizes, and a certificate plan for the TLS and OPC UA links.
Edge Computing and Resilience	Distinction	The outage panel gives the store-and-forward mechanism, what keeps running, the recovery order and the sequence-number check. It holds the band narrowly, because it never sizes the buffer for a long outage or says how the catch-up upload is throttled.	A tiered edge with defined processing at each tier, cost and power figures for the gateway's share of the UPS, and a firmware update plan.
Industry 4.0 Pillar Mapping	Distinction	Six pillars are marked on the layers where they do their work, and the pillar panel follows the AGV arrival times through five of them to a routing decision.	A maturity model running from current state to target, and pillars chosen for the site, such as additive manufacturing for spares.
Visual Design and Communication	Distinction	The flow runs one way, each layer has its own colour and icon, a legend explains the protocol letters and signal steps, a summary box speaks to the plant manager, and red is never set against green.	The right-hand panels are dense. Cutting their text by a third and letting the diagram make more of the argument would move it towards the infographic quality Stretch describes.

Where the figures come from

Every model or log figure on the poster and in this note is read from the W11 sources: the builder, `scripts/build_a04_exemplar_poster.py`, reads each one from these sources, and

`tests/test_assessment_exemplars.py` checks the poster's text against this table. The design choices (protocols, the recovery order, the buffer) are the author's and have no source.

Figure	Value	Source
Cobot speed, mm/s	400	`lagos.json:as_proposed_speed_mm_s`
Required separation S_p, mm	1700	`scenarios.json[current]:separation_required_mm`
Modelled shift, hours	4	`lagos.json:shift_h`
Model protective stops	11	`scenarios.json[current]:stops`
Model trays	469	`scenarios.json[current]:trays`
Model AGV interval, min	25	`lagos.json:agv.interval_min`
Model AGV stops	9	`scenarios.json[current]:agv_stops`
Model restart, s	15	`lagos.json:restart_s`
Separated route: protective stops	2	`scenarios.json[separated]:stops`
Separated route: trays	475	`scenarios.json[separated]:trays`
Log protective stops	22	`shift_log:sum(protective_stops)`
Log trays released	448	`shift_log:sum(trays_released)`
Log AGV arrivals	12	`shift_log:sum(agv_arrivals)`
Log AGV stops	12	`shift_log:sum(stops_agv)`
Log nuisance trips	8	`shift_log:sum(stops_nuisance)`
Log mean restart, s	34.0	`shift_log:weighted_mean(mean_restart_s,protective_stops)`
Log bin width, min	15	`shift_log:bin_minutes`